Diversity, Taxonomic versus Functional

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Glossary

Functional diversity A component of biological diversity that enumerates the different types of processes in a community that are important to its structure and dynamic stability.

Functional group A group of species that utilize similar resources; synonymous with *guild*.

Guild A group of species that utilize similar resources (usually food).

Species evenness A measure of the relative abundance of species in an area.

Species richness The number of species in an area.

Species turnover The change in the composition of species in an area due to the extinction of some species and the replacement by a new species through colonization.

Taxonomic diversity The number and the relative abundance of species in a community.

Species, Communities, and Ecosystems

Linnaeus (1735) and Hutchinson (1959) provided a convenient starting point to address biodiversity in terms of taxonomic and functional diversity. Linnaeus formalized the two-kingdom scheme of Plantae and Animalia and made the distinction between sessile photosynthetic plants and motile food-ingesting animals. Predating an understanding of photosynthesis or modern evolutionary theory, this work was descriptive and simply attempted to categorize and count organisms rather than posit mechanisms. Function was at the basis of the classification. In his address to the Society of American Naturalists, Hutchinson, like Linnaeus, made a deliberate distinction between primary producers and heterotrophs by asking ''Why are there so many kinds of animals?'' The unique feature of this address was that it studied biodiversity by invoking principles from ecology and evolution, formalizing niche theory, and including energetics. More important, it marks an early attempt to link species diversity to the functioning of an ecosystem.

The choice of taxonomic versus functional diversity illustrates a broader issue in science. In both cases, traditional taxonomies based on individual traits or ones based on the functional attributes of a species in a community or ecosystem are human constructions. Ehrlich and Holm (1962) summarized the dilemma nicely:

There seems to be no theoretical reason why there must be complete congruence among estimates of relationships based on characters from different developmental stages or on characters from different organs systems of the same stage.... In a taxonomy based on ecological requirements, whales will be more closely related to sharks than to bears. Such a relationship is no more or less ''true'' than the classical one: it is merely based on different attributes.

At some point, the question comes down to the type of information that is used, the way it is used, and why it is used. The transition from phylogenies based on taxonomies of morphological characteristics to ones based on taxonomies of genetic information have revealed ecological and functional coherence that Ehrlich and Holm (1962) envisioned for higher taxonomic rankings (phyla, orders, and families), if not for species.

Different Approaches to Studying Biodiversity

There are different subdisciplines within ecology: autoecology (individual and population ecology), community ecology, and ecosystem ecology. The focus and approach of each subdiscipline and the type of information each uses have influenced the degree to which taxonomic or functional diversity have been used (**Table 1**).

Individual and Population Approaches

Autecology is the study of individual organisms. The approach originally focused on the adaptiveness of an organism's physiology to the environment but has since been expanded to include the study of the distribution and dynamics of populations. In terms of biodiversity, autecology has embraced taxonomic diversity.

The Community Approach

The community approach defines diversity in terms of the number of species and relative abundances of species. This information is used to study the effects of the number and types of interactions and the consequences that the interactions have on the distributions and abundances of the species and the dynamic stability of the community. Community ecology tends to focus more on taxonomic diversity than on functional diversity. An ideal description of a community would include all species and species interactions. In practice, descriptions of communities include a mixture of individual species and aggregations of species. The aggregations have been based on functional attributes defined by similarities in resource utilization among species and on taxonomic affinities (e.g., ''mammals'').

The Ecosystem Approach

The ecosystem approach defines diversity in terms of processes and function within the community. The focus is not on species *per se* but on the products of species interactions, usually different forms of carbon or nitrogen. Underlying this approach is the belief that species interact in ways that reinforce stabilizing feedbacks that operate to maintain a persistent steady state. The feedbacks operate through the availability of nutrients in one pool that are critical to another pool.

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Table 1 Information used by different subdisciplines of ecology

	Subdiscipline		
	Autecology	*Community ecology*	*Ecosystem ecology*
Focus	Individuals and populations Species	Populations Species and guilds	Forms of matter Functional groups
Units	Individuals	Individuals	Biomass
Biodiversity	Taxonomic	Taxonomic and functional	Functional

A typical description of an ecosystem does not include many species (except for the dominant species) but rather aggregates of species that are similar in the ways in which they process matter relative to a key process. For example, the decomposition of pine needles on a forest floor is a complex process that is influenced by the nutrient quality of the pine needles, the temperature and moisture of the forest floor, and the organisms that are present. For the microbes alone, the process might involve more than 100 species of saprobic fungi and an equal number of strains of saprobic bacteria. Depending on the level of resolution desired, the microbes could be aggregated into a single pool of *decomposers* or separated into pools of *fungi* and *bacteria*. The justification for the seemingly casual treatment of the individual species is that the focus is not on how many species are present but on the general composition of the pool and the rate at which the pool processes matter.

Species, Function, and the Ecological Niche

Taxonomic and functional diversity use similar information but in different ways. The species is central to taxonomic diversity. Species are arranged into phylogenetic relationships that are based on how they acquire energy, their cell types, the level of cellular organization, their embryology and life history, their physiology, and their genetic makeup (RNA or DNA or both). Different classification schemes have been proposed that use all or some of these criteria with different weights given to any one of the criteria. Ecosystem processes are central to functional diversity. Functional diversity has been based on ecological characteristics. Species are aggregated into functional units based on how they use resources. Various schemes have been proposed to aggregate species into functional groupings. Species are arranged into functional relationships based on how they acquire energy, where they acquire energy, the rates at which they process energy (physiology), and when they process energy (life history).

Linking taxonomic and functional diversity would require integrating information that describes the makeup of an organism (its taxonomic characteristics) along with the rates at which it processes and transforms matter (its functional characteristics). Niche theory offers a way to link taxonomic and functional diversity.

The Ecological Niche

The ecological niche of a species has been defined by the resources it requires and by its role or function in the environment. The concept of the ecological niche when introduced by

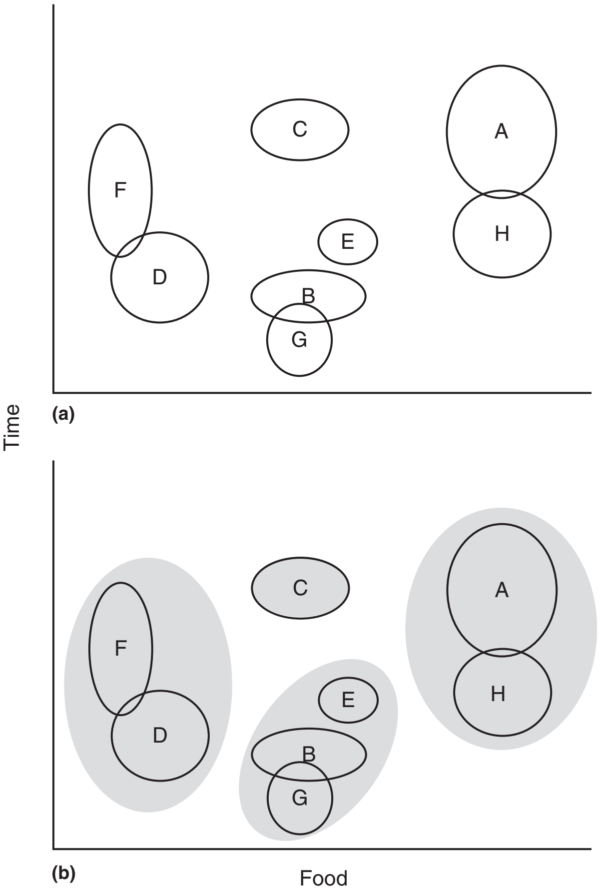

Figure 1 Projection of eight species (A–H) onto two of the principal niche axes (food and time). (a) The ovals surrounding each species represent the region of the food and time space that each species occupies. Ovals that overlap one another represent regions in which the species share (compete for) resources. (b) Ecologists may aggregate species that occupy similar niche spaces into functional groups (species within the shaded ovals).

Grinnell (1917) focused on the habitat requirements of a species for it to survive and reproduce. Hutchinson (1959) expanded on this definition of niche by disaggregating the habitat into the multiple resources it embodied. The physical space a species occupies, the temperature and moisture conditions of the space, and the seasonality in abiotic and biotic conditions that the space experiences, along with the food requirements and the interactions that a species engages in with other species are considered. Here the niche of a species can be viewed graphically as an n-dimensional hypervolume in which each resource represents an independent axis (**Figure 1**).

The space that each species occupies within this hypervolume is defined by its basic resource requirements (i.e., the fundamental niche) and is winnowed down by antagonistic interactions with other organisms (i.e., realized niche). Schoener (1974) presented an analysis of the resource requirements for several species from different taxa, revealing that the principal niche axes are food, habitat use, and time.

Elton (1927) defined the ecological niche in terms of the role of a species in the environment. This definition is subjective but not at odds with resource-based definitions offered by Grinnell (1917) and Hutchinson (1959); rather, it is conditioned on them. The role, or in keeping with this article, function of a species is a property that emerges from the adaptations of species in meeting their resource requirements to survive and reproduce. As for information, both the requirements-based and functional-based definitions of niche are grounded in the genetics of the species, providing a means to link species diversity and functional diversity.

Functional Groupings of Species

Both the community and the ecosystem perspectives have grouped species into larger units when studying interactions among species or processes mediated by species. The schemes used have ranged from those of convenience, which are usually based on a broad taxonomic category, to ones that have adopted strict criteria that may include organisms from different taxonomic groupings. For this discussion, the groupings that have been based on the resource requirements of a species and its position in niche space relative to other species makes the most sense, given that function in an ecosystem context is directly related to how species utilize resources. Hence, a functional group is composed of species that utilize similar resources (i.e., the resource-based niche definition) and perform similar roles (i.e., the functional-based niche definition) in the environment (**Figure 1**). There is no standard number of resources (axes in niche space) used to establish functional groups, but many are based on the principal resources of food, habitat, and time. This has resulted in the groupings being given different names. For example, a *guild* is defined as a group of species that feed on similar food in a similar way. Soil ecologists define a *functional group* as a group of species that feed on similar food in a similar way, occupy the same habitat, and possess similar life histories.

Estimating Diversity

Procedures to estimate species diversity originated from the fields of information theory and statistics. If presented a code with a set number of symbols (s) of known proportions (p), it is desirable to estimate its information content. Ecologists simply substituted the species (or aggregations of species) for the symbols. Species diversity of a community could be estimated in the same way as the information content of a code. Two aspects of species diversity are the number of species (species richness) and the relative abundance of species in a community (species evenness).

Species Richness

Species richness (S) is the number of species within a defined region. The species richness of a region is obtained through sampling or via a census. Because "region" is defined by the observer, species richness has been further categorized into three components to account for changes in spatial scale.

Alpha diversity, sometimes referred to as *point diversity*, is the species richness that occurs within a given area within a region that is smaller than the entire distribution of the species. *Beta diversity* is the rate at which species richness increases as one moves in a straight line across a region from one habitat to another habitat. In other words, it is the rate of change in species richness that occurs with a change in spatial scale. *Gamma diversity* is the species richness within an entire region. As the area being surveyed approaches that of the entire region, alpha diversity approaches gamma diversity and beta diversity approaches zero.

Species Evenness

Species evenness takes into account the number of species and the relative abundance of species in a community. Several indices have been proposed. Two of the commonly used measures of evenness are the Shannon index (H) and the Simpson index (D).

The Shannon index (H) is a measure of the information content of a community rather than of the particular species that is present. The index is as follows:

$$H = -\sum_{i=1}^{S} p_i \log p_i$$

where s is the number of species in the community (species richness) and p_i is the relative proportion of species i. Hence, if two communities (A and B) with the same number of species (not necessarily the same species) present were compared, and the distributions of p_i were the same, then $H_A = H_B$. If either the number of species or the proportions differed, however, then $H_A \neq H_B$. In the latter case, the community with the greater number of species whose relative abundances were equal would possess the higher diversity.

The Simpson index (D) measures the dominance of a multispecies community and can be thought of as the probability that two individuals selected from a community will be of the same species. The Simpson index was originally proposed as follows:

$$D = -\log \sum_{i=1}^{s} p_i^2$$

where S is the species richness of a community and p_i is the relative proportion of species i. The index can be modified to 1–D to give it the property of increasing as diversity increases (the dominance of a few species decreases).

Linking Species Richness and Species Evenness

Species richness and species evenness are special cases of dichotomous-type rarity measures. For dichotomous-type rarity measures for a pool of s species, the rarity of species i, whose function is denoted $R(p_i)$, depends only on its relative abundance (p_i). The rarity measures (Δ) take the following form:

$$\Delta = \sum_{i=1}^{s} p_i R(p_i)$$

The relationships between species richness (S), the Shannon index (H), and the Simpson index (D) are presented in Table 2. Notice that the indices share three important properties: (1) Δ is maximized when $p_i = 1/S$ for all i species; (2) given that $p_i = 1/S$, the community with the higher species richness has the greater diversity; and (3) $\Delta = 0$ if the community possesses a single species ($S = 1$).

Biodiversity and Systems Theory

One of the enduring tenets of ecology in the early part of the 20th century was that a diverse community was a stable community. The linkage between biodiversity and stability has an intuitive appeal. The more species there are in a community, the more likely they would be tightly networked, and hence the more stable the community would be. Interestingly, function plays an important role in this argument in that in a highly diverse community, there would be alternative species or assemblages of species to perform key tasks to sustain the cycling of matter and the current diversity. These ideas have been challenged and modified. Ecologists agree that biodiversity is important to the function and stability of an ecosystem. The current debates concerns the nature of the relationship between biodiversity, function, and stability.

Table 2 Rarity functions and forms of diversity indices

	Rarity function	*Diversity index*
Species richness (S)	$R(p_i) = (1/p_i) - 1$	$\Delta = S - 1$
Shannon index (H)	$R(p_i) = -\log p_i$	$\Delta = -\sum p_i \log p_i$
Simpson index ($1 - D$)	$R(p_i) = 1 - p_i$	$\Delta = \sum p_i(1 - p_i)$

Although useful, the functions used to estimate biodiversity that were presented previously are descriptive measures and imply neither the mechanisms that shape biodiversity nor the importance of biodiversity to maintaining the structure and dynamic stability of a community. If the diversity of one community were higher than that of another, does this increase its likelihood of persisting in time or recovering from a disturbance? The answer depends on the species involved. For example, the colonization of an exotic species increases the species richness of a community but does not necessarily increase its stability – to the contrary, it often destabilizes the system. The addition of a predator, however, can stabilize an ecosystem by tempering the oscillations in the population densities of its prey through time. Biodiversity is a property of complex dynamic biological systems. The steady state and stability are two concepts important to the study of complex dynamic systems.

The Steady State

A steady state is an equilibrial condition in a dynamic process in which the rate of input of a state variable is equal to the rate of output of that variable (Figure 2). The dynamic process can involve any variable that changes in time. For taxonomic and functional diversity, the variables of interest are the number of species or functional groups, the densities and biomass of each species or functional group, and the mass of key elements within different pools within the ecosystem.

The Number of Species

The number of species in a community is at steady state when the rate of colonization equals the rate of extinction (Figure 3). The classic model of this process, the theory of island biogeography, assumes that the rate of colonization is affected by

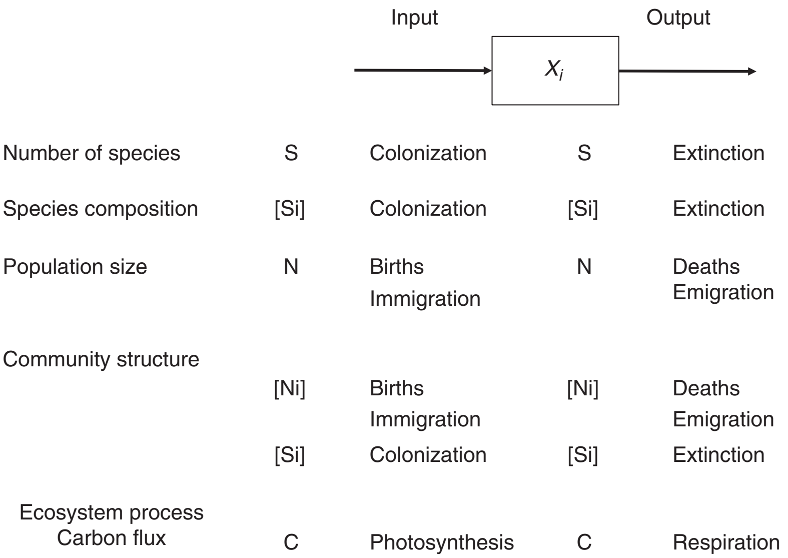

Figure 2 Diagram and list of the processes (inputs and outputs) that contribute to the dynamics of a single state variable (X) or several state variables [X_i].

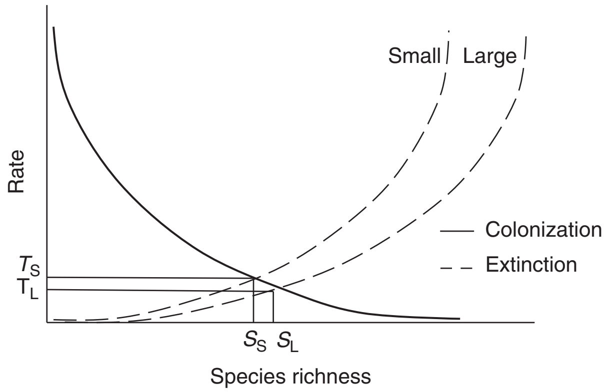

Figure 3 The effect of the rates of colonization and extinction on the species richness of large and small islands (S_L and S_S, respectively). T_L and T_S represent the points at which the rates of colonization and extinction are equal. Adapted from MacArthur RH and Wilson EO (1967) *The Theory of Island Biogeography*, *Monographs in Population Biology*, vol. 1. Princeton, NJ: Princeton University Press.

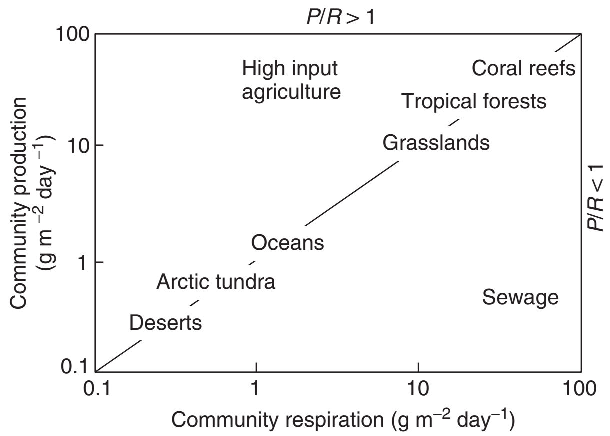

Figure 4 The relationship of various ecosystems in terms of their community production (grams per square meter per day), *P*, and community respiration (grams per square meter per day), *R*. The dashed line represents the steady state in terms of *P* and *R* ($P/R=1$). Note the position of the disturbed ecosystems.

how far the habitat being colonized is from the source of species and by the size of the habitat being colonized. Colonization rates are higher for large habitats that are close to the source of species than for smaller distant habitats. The rate of extinction is affected by the size of the habitat. The rates of extinction for smaller habitats are higher than those for larger habitats.

Species Composition

In the model presented previously, although the number of species is constant at steady state, the species composition of the community does not remain constant. The model simply assumes that once the rates of colonization and extinction are equal, then for every new species that enters a habitat, one of the current resident species goes extinct. This point is important when trying to estimate the diversity of a habitat by sampling the habitat over time. If a habitat were sampled long enough, the estimate of the number of species within the habitat would equal the number of species in the habitat serving as the source, even though the source has a higher number of species at its steady state than the habitat being colonized.

Population Densities

The population density of an individual species within a habitat is at steady state when the number of births and immigration equal the number of deaths and emigration. When modeling this process, ecologists typically assume that the habitat is closed to immigration and emigration and focus entirely on the rates of birth and death. More sophisticated approaches that involve multiple habitats will include the immigration and emigration of species among habitats.

Community Structure

A community is at steady state when the species composition of the community and the densities of the species remain constant. This implies that the species richness and evenness of the community do not change. This condition is rarely met since the species composition and the densities of the species fluctuate through time.

Ecosystem Processes

An ecosystem process is at steady state when the rate of input of matter or energy is equal to the rate of output. The input of matter and energy occurs via photosynthesis and from sources external to the ecosystem (e.g., leaves entering a stream). The export of matter and energy occurs through heterotrophic respiration and the physical movement of matter from the ecosystem (e.g., export of leaves from forest to a stream). The production of organic matter from inorganic matter (photosynthesis) and the transformation of organic matter to other organic forms and inorganic matter (heterotrophic respiration, hereafter referred to as *respiration*) are ecosystem processes that approach a steady state and illustrate the interplay between matter and energy. When the rate of photosynthesis equals the rate of respiration, the system is said to be at steady state (**Figure 4**). In other words, at steady state the ratio of production to respiration for matter and for energy is 1 ($P/R=1$).

Stability

A population, community, or ecosystem is stable if it can persist through time and can recover from minor disturbances. A disturbance is any perturbation to the system that causes any of the species or functional groups to deviate from their steady state. There are many facets to stability and many operational definitions as well. Aspects of stability include local asymptotic stability and short-term transient dynamics.

Local Asymptotic Stability

An ecosystem possesses local asymptotic stability if over time it returns to its original steady state following a minor disturbance. The minor disturbance is relative to the sizes of the populations involved and occurs in the neighborhood of the steady state, hence the reference to "local." This concept of stability has been widely used by biologists because it is simple and exact in its interpretation. It has been criticized for the same reasons that it has been accepted. For example, high species diversity has been shown to negatively impact the local stability of model ecosystems. This result was the basis for the hypothesis that dynamic stability constrains biodiversity. However, competing models of ecosystems with different sets of assumptions and criteria for stability, although they do not disagree that dynamic stability affects biodiversity, yield different results with regard to the nature of the relationship.

The concepts of local stability and the theory of island biogeography may seem at odds with one another, even though they rely on similar mathematics and assumptions and are modeling the same systems. An obvious question is, how can an ecosystem be locally stable if the species composition of the ecosystem is changing? The discrepancy can be clarified by revisiting the processes that are being modeled. When modeling populations within a community, local stability is in reference to the densities or biomass of the populations. The dynamics of the populations are governed by the difference between the rates of birth and death. When modeling the species diversity within the same community, local stability is in reference to the number of species that are present. The dynamics of the number of species is governed by the rates of colonization and extinction. To carry this a step further, the local stability of the carbon or nitrogen cycles within an ecosystem is in reference to the mass of the different forms of carbon or nitrogen. The dynamics of the different forms is governed by the functional attributes of the species present and the rates at which the different forms are acted on.

Transient Dynamic Behavior

Transient dynamics represent the short-term dynamic behavior that asymptotically stable systems exhibit following a minor disturbance. The nature of and degree of the transient response can influence the stability and diversity of the system. Transient behavior includes the amplification and decay of a perturbation, represented by an amplification envelope, $\rho(t)$, encompassing the magnitude and extent of the deviation and its temporal components (**Figure 5**). Following a disturbance, the perturbation to the system, now portrayed as deviations from equilibrium, amplify to a maximum ρ_{max} at time $T_{max,}$ with a rate defined by its reactivity during the short-term transient phase.

Eventually, the perturbation decays to the long-term asymptotic phase at a time defined by its return time at a rate defined by its resilience. The principle of resilience can be applied to any state variable present in an ecosystem: population densities, the number of species, or the mass of nutrients. Resilience is a function of the rates of inputs of the state variables, the rate at which the state variables turn over, and the number of state variables. The return time of a system is a measure of its resilience and is estimated as the time between the point when the disturbance altered the ecosystem and the point at which the ecosystem recovers. More resilient ecosystems have shorter return times than less-resilient ones.

The persistence of a system and hence its diversity may be compromised if the system possesses a highly reactive transient phase or if the frequency of disturbance is too great given the length of return time because a system may not have time to recover from any one incident. Under these circumstances, the perturbations may grow initially and the populations may oscillate wherein the densities of individual species may approach zero, making them vulnerable to future disturbances, or some may dip below a lower threshold where the likelihood of successful reproduction is compromised. Frequent disturbances of reactive systems may also push the system and its dynamic state from one state and attractor to another. Again, although there is uncertainty of the effects of this type of transition on the system, a consequence is the loss of species or the introduction of new species.

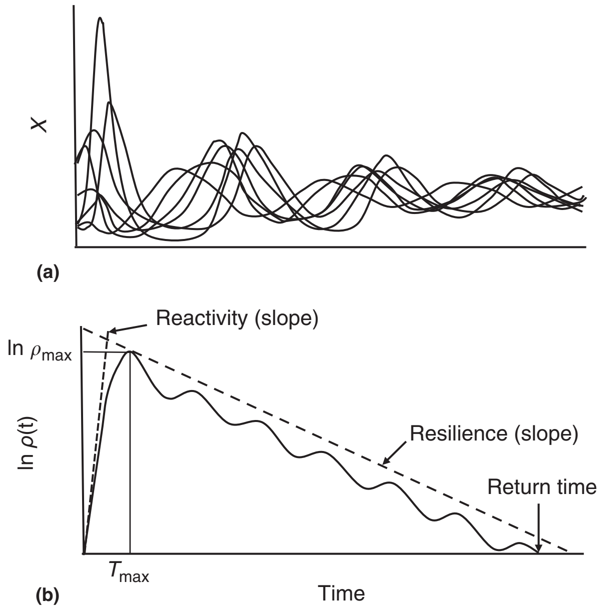

Figure 5 An overview of transient dynamic behavior. (a) The transient dynamics of a multispecies system that has been disturbed as it returns to equilibrium. (b) A summary of the components of the transient behavior of a reactive system in terms of the amplification envelope $\rho(t)$ through time. The deviations from equilibrium amplify to a maximum ρ_{max} at time T_{max} with a rate defined by its reactivity (slope of —) during the short-term transient phase. The disturbance eventually decays at a rate defined by its resilience (slope of - - - -) to the long-term asymptotic phase at a time defined by its return-time. Adapted from Chen XIN and Cohen JE (2001) Global stability, local stability, and permanence in model food webs. *Journal of Theoretical Biology* 212: 223–235.

Modeling Species Interactions

A simple mathematical exercise illustrates how the number of interacting units (species) within a system and the number of interactions between these units (species interactions) are related to the stability of the system. The exercise constructed simple systems of four, seven, and 10 units assumed to be at a steady state. For a system of a given size, connections are made between the units in a way that ensures that all units would be connected to the system, creating several variations of the system in terms of the connections. Next, the strength of the connections is allowed to vary, and the systems are disturbed by moving the units away from their steady state. Two results emerge from this exercise. First, as the number of interacting units increase within a system the likelihood of the system returning to a steady state after a small disturbance (stability) decreases. Second, for a given number of interacting units, as the number of connections between the units increases, the stability of the system declines.

There is a clear connection between the simple mathematical exercise previously presented for estimating diversity and the diversity and stability of ecological communities. The species is substituted for the generic units modeled previously, and the interactions are defined in terms of predation, competition, or mutualism. Like the generic modeling exercise, high biodiversity does not necessarily lead to stability. Additional study demonstrates that systems composed of species aggregated into blocks of highly interactive species that are connected by weaker interactions are more stable than systems with the same number and types of species and interactions arranged at random. Hence, the way in which the community is constructed affects its diversity and stability.

The types of interactions among species affect the stability of a community as well. Theory and observation agree in general terms on this point, if not on specifics. For example, mutualistic interactions, whereby the actions of each species benefit the growth and dynamics of one another, and omnivory, in which species derive energy from different trophic levels, have been shown to be destabilizing in simple models, but both are prolific in natural communities. All the dominant terrestrial ecosystems rely on a mutualism between the vascular plants and microbes, and nearly every described ecosystem possesses numerous examples of omnivory.

Modeling Species and Processes

Process-oriented models focus on the transformation of elements among various inorganic to organic states through ecosystems. Carbon, nitrogen, and phosphorus are the more common choices of elements to be modeled given their importance to plant and animal growth. Models of species interactions have been connected to models of processes. The general approach to linking these models is to use the structure and formulation adopted when modeling species with the currency used in models of processes. First, species are aggregated into functional groups using one of the schemes described previously, and the interactions among the functional groupings are established. Second, the biomass (expressed in terms of carbon, nitrogen, or phosphorus) of each functional group is estimated. Third, a series of differential equations is developed that establishes which functional groups interact with one another and the nature of the interactions. Parameters within the models include birth and death rates, feeding rates, and the efficiencies by which the functional groups convert consumed matter into new biomass. The models are used to estimate the flow of elements among the functional groups, the flow of elements out of the system, the stability of the ecosystem, and the resilience of the ecosystem.

The following results have emerged from these models: First, the models are sensitive to the changes in the parameters associated with the physiology and life histories of the species within the functional groups. For example, microbes possess high reproductive rates and high rates of energy conversion, and they process matter at a higher rate compared to a functional group dominated by arthropods or nematodes, which possess lower reproductive rates and rates of energy conversion. Second, functional groups are arranged into interactive blocks that form pathways of material flow that originate from a specific source and end with a top predator. The blocked arrangement of functional groups, the distribution of biomass within the ecosystem, and the distribution of the flow of elements through the ecosystem influence the stability and resilience of the system.

Linking Taxonomic and Functional Diversity

The linkage between taxonomic diversity and functional diversity involves many facets. The morphology of an organism determines the type of habitat in which it resides and its ability to colonize new habitats. The physiology of a species influences its adaptiveness to a habitat and the rates and efficiency with which it transforms matter. The life history of an organism influences the rate at which it processes matter (births, deaths, and consumption). Linking taxonomic diversity to functional diversity requires finding measurable attributes that all organisms possess that have meaning to all the ecological subdiscplines, but it also requires flexibility in terms of how the information is used. Three case studies are presented in the following sections. The case studies illustrate how, when studying biodiversity, aspects of both taxonomic and functional diversity are used interchangeably.

Diversity of C3 and C4 Grasses

The first case study involves a comparison of the distribution of grass species that possess either the C3 or C4 photosynthetic pathways in North America. The study illustrates how the information of autecology can be used to find mechanisms to explain patterns in diversity. The study demonstrates how the functional aspects of an organism's physiology influence a species adaptation to a habitat and ultimately the diversity of species with similar physiologies.

Plants that possess C3 and C4 photosynthetic pathways differ in terms of their morphology and biochemistry as they relate to the dark reaction of photosynthesis. For C3 plants, photosynthesis occurs entirely within the outer mesophyll cells, in which CO_2 enters the Calvin–Benson cycle through the plants' stomata. The enzyme (RuBP carboxylase or Rubisco) that incorporates CO_2 into the Calvin–Benson cycle also has an affinity toward O_2. If O_2 were incorporated instead of CO_2, the plant would experience photorespiration – energy is expended, no carbon is fixed, and the plant loses organic carbon. For C4 plants, there are two types of photosynthetic cells: the outer mesophyll cells and the inner bundle sheath cells. In C4 plants, CO_2 enters the mesophyll through the stomata and is incorporated into a four-carbon compound before being exported to the bundle sheath cells in which it enters the Calvin–Benson cycle. The additional steps that occur within the C4 plants prior to the Calvin–Benson cycle cost the plant additional energy and organic carbon. However, the enzyme (PEP carboxylase) that adds the CO_2 to the four-carbon molecule in the mesophyll cells is more efficient than Rubisco and less likely to initiate the loss of organic carbon to photorespiration. This trade-off places plants with the C4 pathway at a disadvantage under cool and moist conditions compared to those with the C3 pathway, but it affords them an advantage in warmer and dry conditions.

Within North America, the northern latitudes are dominated by species of C3 grasses, whereas the southern latitudes are dominated by species of C4 grasses. This pattern of

Table 3 Functional groups of arthropods by number and relative proportion (%) on islands before and 1 year after fumigation

	Number and relative proportion (%)							
	Herbivore	*Scavenger*	*Detritivore*	*Wood borer*	*Ant*	*Predator*	*Parasite*	*Total*
Before	55 (34)	7 (4)	13 (8)	8 (5)	32 (20)	36 (22)	12 (7)	163
After	55 (40)	5 (4)	8 (6)	6 (4)	23 (17)	31 (23)	9 (6)	137

Source: Adapted from Heatwole H and Levins R (1972) Trophic structure, stability and faunal change during colonization. *Ecology* 53: 531–534.

diversity has been attributed to the adaptiveness of each pathway to patterns of temperature and light intensity. The lower latitudes possess higher temperature and higher light intensity than the upper latitudes given the orientation of Earth to the sun. Grasses that possess the C3 pathway have been shown to be more productive than those with the C4 pathway under conditions of cool temperature and low light intensity. At higher temperatures and light intensity, C3 plants suffer greater water loss through their stomata and increased photorespiration than do C4 plants.

Test of the Theory of Island Biogeography

This case study illustrates how information from the community perspective has been used to study biodiversity. The experimental manipulations of several small mangrove islands off the Florida coast by Simberloff and Wilson (1969) and the reanalysis of their work by Heatwole and Levins (1972) offer not only the best validation of the theory of island biogeography but also some insight into how taxonomic diversity and functional diversity are related.

The arthropods on four small mangrove islands were surveyed. The islands were then fumigated with methyl bromide to eliminate the arthropods. For 2 years following the disturbance, the islands were resurveyed for arthropods to capture the recolonization process. The results were consistent with those predicted by the model proposed for the theory of island biogeography (**Figure 3**). The colonization rates were higher and the steady states in species richness were achieved sooner on islands closer to the mainland than on the more distant island. Once a steady state in species richness was achieved, the species composition changed due to the predicted turnover in species.

The study provides an explicit link between the recovery process, species diversity, and functional diversity when the colonization of functional guilds – groups of species that share common food and foraging behaviors – is compared to that of the individual species. One year following the fumigation, the taxonomic diversity of the islands was lower than it was prior to the fumigation (**Table 3**). However, the proportion of species within each of the guilds was the same as it was prior to fumigation. These results suggest that the functional diversity of the islands recovered before the species diversity. In other words, the taxonomic diversity of the communities was in some way dependent on functional characteristics of the community.

Invasion of an Exotic Species

The final case study illustrates how elements of taxonomic and functional diversity and information from autecology, community ecology, and ecosystem ecology have been used to study biodiversity. Vitousek *et al.* (1987) studied the natural history of the invasive plant species *Myrica faya* and the effects that it has had on the nitrogen cycle and plant community development of recent lava flows in Hawaii. Since its introduction to Hawaii from the Canary Islands by Portuguese immigrants at the end of the 19th century, *M. faya* had colonized more than 34,365 ha. *Myrica faya* forms a symbiotic association with the soil nitrogen-fixing actinomycete, *Frankia*. The plant possesses two characteristics that separate it from the native nitrogen-fixing plants on the islands. Unlike the native plant nitrogen fixers (*Acacia koa* and *Sophora chrysophylla*), the seeds of *M. faya* are small, have an edible fleshy fruit, and pass through the digestive systems of birds intact. Moreover, *M. faya* is a more efficient nitrogen fixer than the native species. When combined, these factors make *M. faya* a superior colonizer of recent lava flows, pastures, and forests.

Once established, the *M. faya* and its soil symbiont, *Frankia*, alter the nitrogen status of the soils. For example, soil nitrogen inputs within open canopies of *M. faya* were four times more than those of similar sites without the exotic plant (23.5 and 5.5 kg per ha per year, respectively). The added soil nitrogen was shown to facilitate invasions by other exotic species (e.g., strawberry guava, *Psidium catleianum*) to the detriment of native plant species. The functional attributes of a single species and its symbiont affected the biodiversity of a community by altering the course of community development.

Ecological Coherence of High Taxonomic Ranks

The acquisition and disposition of matter and energy are key processes that all organisms possess that directly link them to ecologically relevant functions. The case studies previously presented share this common feature. The acquisition of matter and energy are characteristics of the interactions in question, and they occur or emerge at taxomomic ranks higher than species. This should not be too surprising given that the modern phylogenies developed from the times of Linnaeus (1735) through Whittaker (1969) relied on energy acquisition and morphology (which is related in many ways to energy acquisition). Recent treatment of prokaryotes that are based on gene sequences have entertained this idea of including environmental information into the calculus of forming taxonomic clades. Assessments of bacterial and archaeal phylogenies when viewed through this lens confirm that there appears to be greater ecological coherence of organisms at high taxonomic ranks. Reducing ecological diversity to functional groups of species that share these attributes may in fact be the way to organize information captured within species to make the connection between functional and species diversity.

Natural Selection

Species evolve and communities and ecosystems change, but do communities and ecosystems evolve? Does natural selection operate on communities and ecosystems even though they do not possess genes? Natural selection needs to be defined before these questions can be answered. Natural selection is the differential perpetuation of genes in successive generations caused by different degrees of adaptiveness to the environment.

One argument is that ecosystems do evolve because the composition of species within a community changes and natural selection operates on the gene pools of each species. In addition to the changes in the genetic makeup of the species, there are changes in key processes that feed back to reinforce stable configurations or that initiate change. A criticism of this analogy is that natural selection at the ecosystem level for the type of coordination among species that leads to a stable ecosystem is not the same as selection at the organismal level on genes. Selection on genes operates through differential reproductive success of individuals, whereas this may not be true at the ecosystem level since the boundaries of an ecosystem are not so clearly defined.

An alternative view posits that ecosystems do not evolve but that the species composition of the stable ecosystem is a result of the selection operating on individuals and the coevolution of species interacting with one another. The outcome of selection on individuals can influence function within a community, which in turn feeds back as a form of selection. Unique to this argument is that the adaptiveness of a species includes the degree to which its dynamics and function contribute to and are compatible with the stability of the ecosystem as a whole.

Appendix

List of Courses

1. Biogeography
2. Ecosystem Ecology
3. Introduction to Ecology

See also: Guilds. Habitat and Niche, Concept of. Island Biogeography. Measurement and Analysis of Biodiversity. Plant Invasions. Species Diversity, Overview. Stability, Concept of. Taxonomy, Methods of

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